

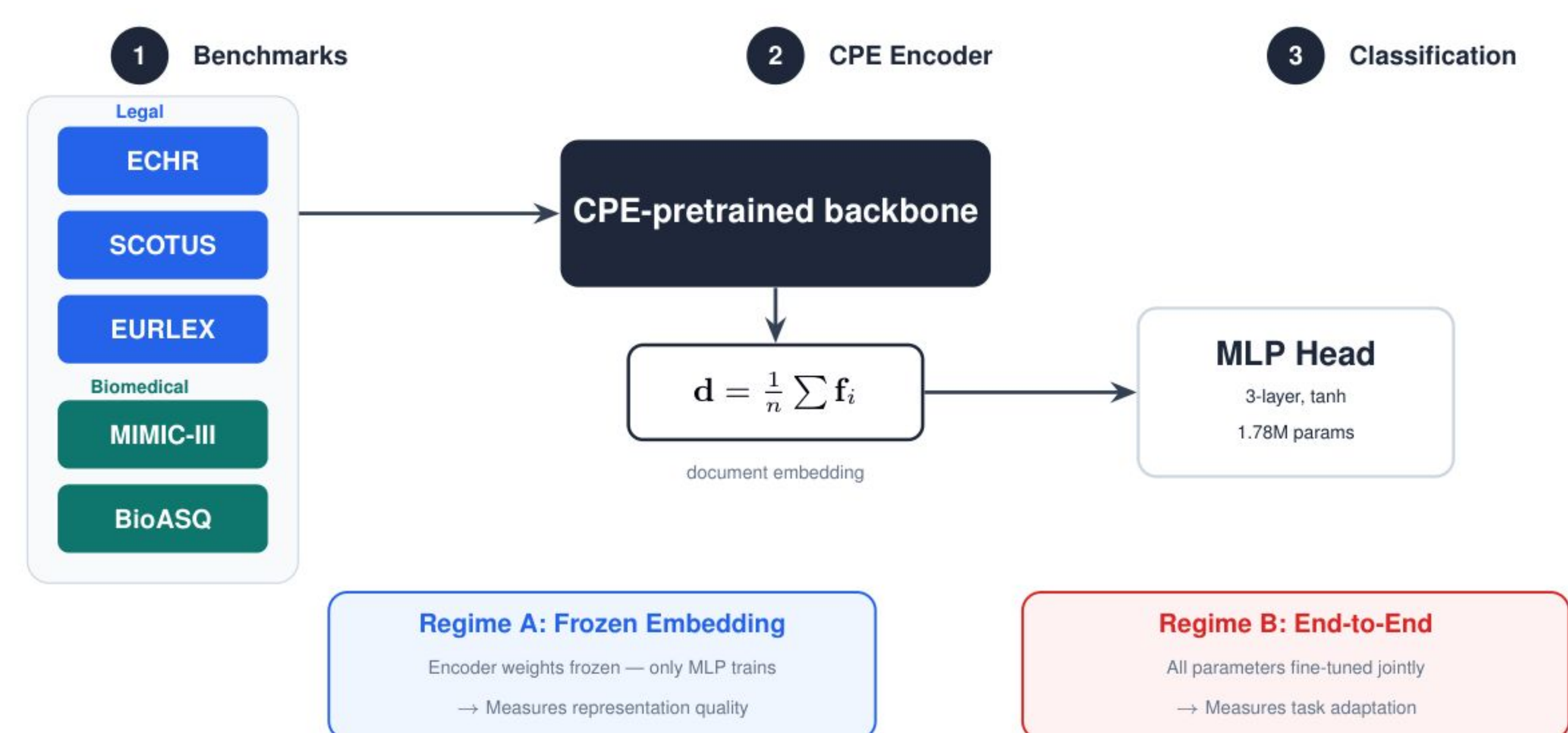
Learning Long-Document Embeddings via Chunk-Context Entailment

Waheed Ahmed Abro - Naïm Es-Sebbani - Zied Bouraoui
CRIL – CNRS & Université d'Artois, Lens, France.

01 INTRODUCTION

Learning faithful embeddings for long documents remains challenging, especially in domains like law and medicine where inputs are long, structured, and semantically heterogeneous. Existing contrastive methods such as SimCSE and ESImCSE focus on sentence-level similarity and fail to capture the hierarchical, directional relations that exist within long documents. When domain experts read lengthy texts, they selectively focus on informative fragments that collectively convey the document's meaning. Inspired by this cognitive process, we propose CPE — a self-supervised framework that frames document understanding as chunk–context entailment, asking: does this fragment belong to this document? CPE combines contrastive learning with a binary entailment head into a joint objective that yields embeddings which are both geometrically smooth and semantically coherent across document sections.

03 Evaluation Setup

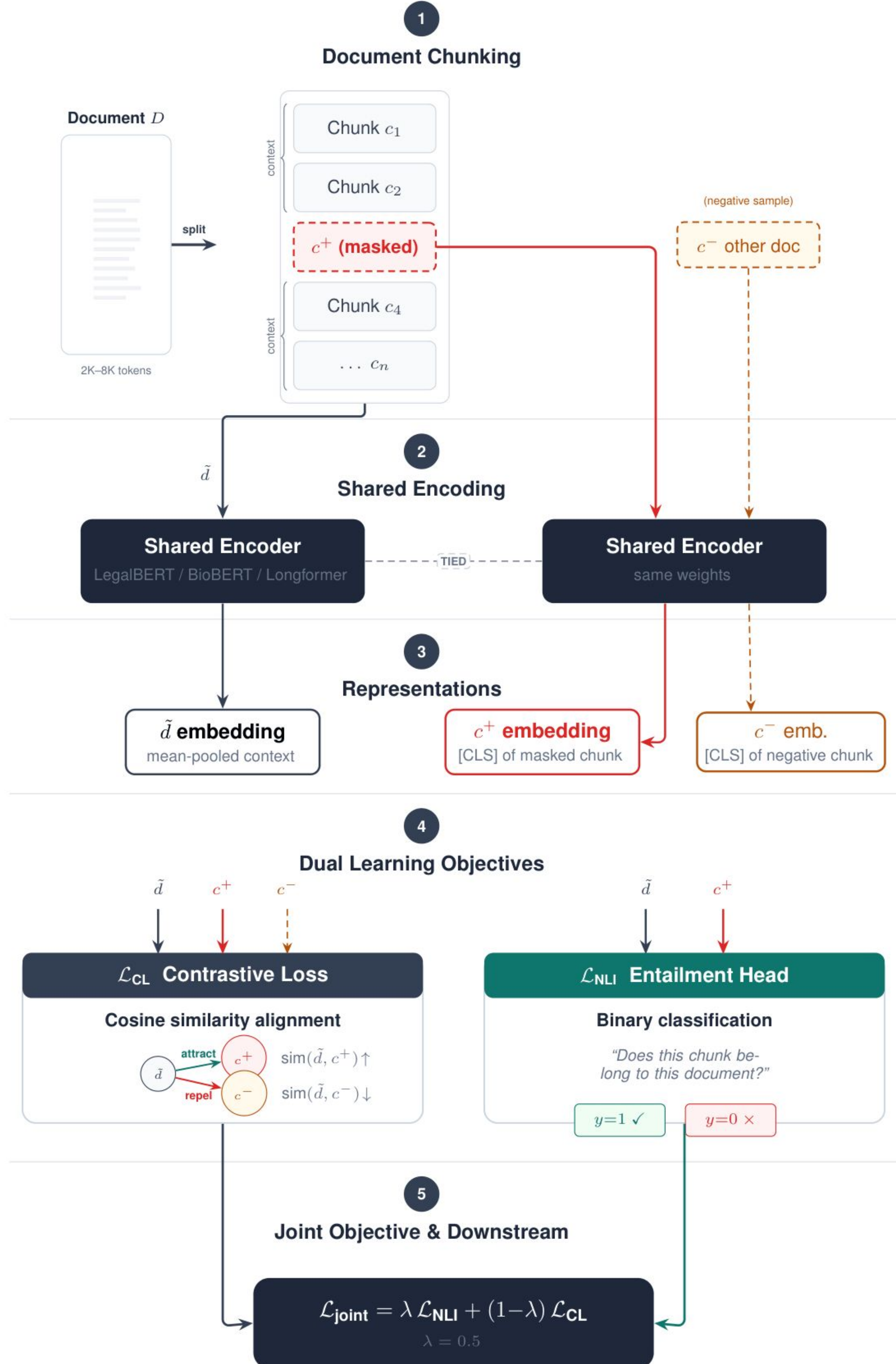


04 Results

PTM	Model	ECHR		SCOTUS		EURLEX		MIMIC	
		Ma	Mi	Ma	Mi	Ma	Mi	Ma	Mi
BERT	Embedding + MLP	36.3	55.6	40.6	61.8	25.9	51.4	49.1	63.0
	SimCSE	48.7	61.4	45.5	61.1	33.4	57.1	54.6	67.1
	ESimCSE	50.8	64.4	52.1	64.1	34.2	54.6	57.0	68.1
	ArcCSE	52.1	65.1	53.6	65.4	36.2	59.0	57.3	68.2
	SimCSE++	52.9	65.7	53.8	65.8	37.4	60.2	57.9	68.5
	DistillCSE	53.2	66.0	54.1	66.0	38.5	61.0	58.0	68.6
	CPE (ours)	54.8	67.0	54.6	66.8	41.0	63.7	59.2	69.1
RoBERTa	Embedding + MLP	35.3	55.6	32.8	57.9	21.9	35.1	49.6	64.3
	SimCSE	49.0	59.4	48.3	62.7	35.1	53.6	55.7	67.0
	ESimCSE	45.3	56.6	57.7	67.9	33.8	54.1	57.5	67.3
	ArcCSE	52.8	64.0	56.1	67.5	37.9	60.8	58.0	68.1
	SimCSE++	54.1	64.9	57.3	67.9	38.8	61.7	59.2	68.8
	DistillCSE	54.6	65.0	57.6	68.0	39.0	62.0	59.3	69.0
	CPE (ours)	56.0	65.6	57.9	68.1	41.9	63.4	61.0	69.4
LegalBERT	Embedding + MLP	52.6	65.3	44.9	65.4	17.8	39.5	—	—
	SimCSE	57.5	68.8	57.6	68.9	40.9	61.6	—	—
	ESimCSE	56.5	68.3	56.1	67.8	40.4	62.6	—	—
	ArcCSE	57.0	68.7	57.9	69.1	41.1	63.0	—	—
	SimCSE++	57.2	69.0	58.2	69.3	41.5	63.3	—	—
	DistillCSE	57.4	69.1	58.6	69.4	41.6	63.5	—	—
	CPE (ours)	57.7	69.4	59.5	71.9	42.2	64.2	—	—
ClinicalBioBERT	Embedding + MLP	Not applicable to legal datasets						55.8	68.9
	SimCSE	Not applicable to legal datasets						62.7	71.1
	ESimCSE	Not applicable to legal datasets						60.4	70.9
	ArcCSE	Not applicable to legal datasets						61.9	71.2
	SimCSE++	Not applicable to legal datasets						62.3	71.4
	DistillCSE	Not applicable to legal datasets						62.9	71.5
	CPE (ours)	Not applicable to legal datasets						63.9	71.7

Classification performance on document embeddings produced by contrastively trained hierarchical transformer models on ECHR, SCOTUS, EURLEX, and MIMIC datasets. Performance is reported in F1 scores $\times 100$ (macro / micro). Best per column in bold.

02 Chunk Prediction Encoder



05 Conclusion

CPE provides a simple, effective, and architecture-agnostic approach to long-document representation learning. By framing chunk–context compatibility as an unsupervised NLI problem, it yields embeddings that are both geometrically smooth and semantically coherent — achieving up to +4 macro-F1 on average across five benchmarks in legal and biomedical domains. CPE also trains faster (3h vs 4.5h on ECHR) and dramatically outperforms zero-shot LLaMA-3 8B (54.8 vs 15.4 macro-F1). Future directions include multilingual extensions, retrieval-augmented applications, and generative document summarization settings.

Scan QR to check more results and visualizations.

